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Studierichting: Informatica
Oefeningenreeks 7A nr. 11.1

$$f : \Omega \subseteq \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \rightarrow z \cdot \operatorname{Bgtan} \frac{x}{y}$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \left(z \cdot \operatorname{Bgtan} \frac{x}{y} \right) = z \cdot \frac{1}{1 + \left(\frac{x}{y} \right)^2} \cdot \frac{1}{y} = \frac{zy}{x^2 + y^2}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left(z \cdot \operatorname{Bgtan} \frac{x}{y} \right) = z \cdot \frac{1 \cdot x}{\left(1 + \left(\frac{x}{y} \right)^2 \right) (-y^2)} = -\frac{xz}{y^2 + x^2}$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} \left(z \cdot \operatorname{Bgtan} \frac{x}{y} \right) = \operatorname{Bgtan} \frac{x}{y}$$

$$Df = \left(\frac{zy}{y^2 + x^2} \quad -\frac{xz}{y^2 + x^2} \quad \operatorname{Bgtan} \frac{x}{y} \right)$$

References